

Porting VPD §4.3: do components interact through the QK circuit?

VPD decomposes $W_{QK} = \sum_{c,c'} \dots$ over subcomponent PAIRS. Their components are rank-1 so that term is a scalar; ours are not, but the decomposition is still exact, and the pair strength generalises to $\|A_{Q,c}^T R_{\tau} A_{K,c'}\|_F = \sqrt{\text{tr}(R^T G_Q R G_K)}$, which reduces to VPD's quantity at rank 1. Measured over all 16×32 heads and 13 RoPE offsets: the interaction is overwhelmingly BETWEEN components, matching VPD's finding that circuits are not contained in one subcomponent — but ablation finds no synergy, so the pairing is not causally load-bearing.

